

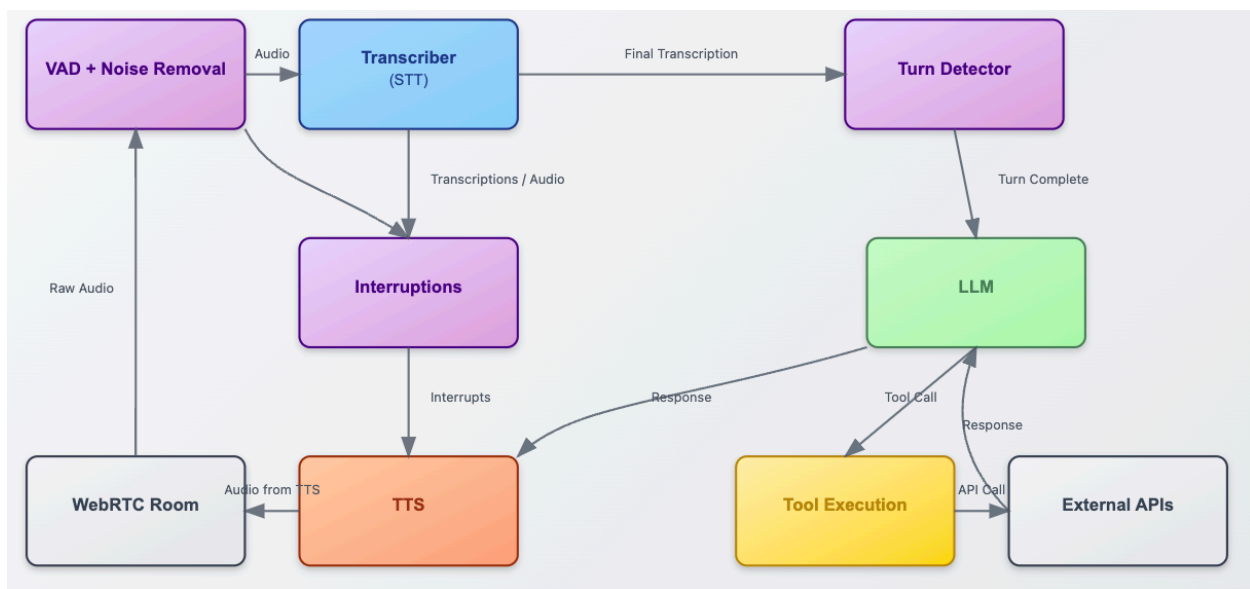


AI / ML Engineering Task (EoT)

Intro

Voice AI agents are composed of multiple models orchestrated together to generate well thought and realistic voice experiences.

Here is a diagram of a common voice AI agent architecture :



If you want to read more about voice ai agents →

<https://voiceaiandvoiceagents.com/#conversational-voice-ai>

Turn detection model

Turn detection means determining when the user is finished talking and expects the LLM to respond.

In the academic literature, various aspects of this problem are referred to as *phrase detection*, *speech segmentation*, *end of sentence*, and *endpointing*. (The fact that there is academic literature about this is a clue that it's a non-trivial problem.)

Our voice agent must quickly detect when it's the agent's turn to talk, without interrupting the user and without adding unnecessary latency.

Your task is to design a smart detector that works :

- For English.
 - Bonus: Make it multi-lingual.
- Is low latency (ideally <100ms per request)
- Is production-ready: trained, evaluated and served (fastapi for example).

The fun part - Task (around 9 hours)

1. Data - 2/3 hours

Create a dataset if needed. You can look online for public datasets, find conversations on youtube, synthesize samples with LLMs ... get creative!

2. Baseline and Model - 2/3 hours

1. Train a small model or fine-tune an existing one. Feel free to use any method you like (e.g. text-based, audio-based, etc.)
2. Evaluate your model.

Be thoughtful about ML best practices (cross-validation, held-out datasets, metrics etc)

3. Serving & Performance - 1/2 hours

1. Expose your model via API (e.g. FastAPI) and build a simple inference script to test it out AND stress test it.
2. Measure throughput / latency on your local machine. Write your findings, concisely and clearly.

Include a Dockerfile.

4. Monitoring - 1 hour

Can you think of a plan to monitor this model in production?

- How would you do it? What are important things we should measure?
- How can we improve this model over time? Are we limited by the datasets previously built?

Write down your ideas and how you would approach this problem if you were working at HappyRobot. We are currently doing almost 1M calls a month and have access to a lot of production data!

5. Discussion - 1 hour

We will have a call to discuss the solution and questions around it. You can take some time to think of the following questions (even write them down and share):

- What do you think are the limits of your current solution, in terms of modeling? In terms of data?
- Do you think an audio-based solution is better than a text-based? Multi-modal? What are the benefits of one or the other? why would they be better than a VAD approach?
- Do you think a transcriber model could handle this task if architected and fine-tuned accordingly? Can you think of a way to do this? You can use Whisper or Parketeer as transcription models to think about this.
- How would you integrate this in a voice agent architecture.

Deliverables

- A short video going over the solution (maximum 10 minutes). No need to go into the details.
- Short document presenting your solution.
- Well documented code.

Side notes

- We know it's a big task. Don't spend too much time optimizing accuracy and modeling (hyperparameter tuning ...). We want to see how you think and approach the problem more than anything else. You will have time with us to explain too.
- Feel free to make assumptions (just write them down). You can also ask clarifying questions, but don't block on them, pick a reasonable path and ship.